



NOAA
FISHERIES
SWFSC



NOAA Fisheries Glider Challenge:

Accelerating the Use of Passive Acoustic-Equipped Gliders to
Monitor and Assess Cetacean Populations in the Pacific

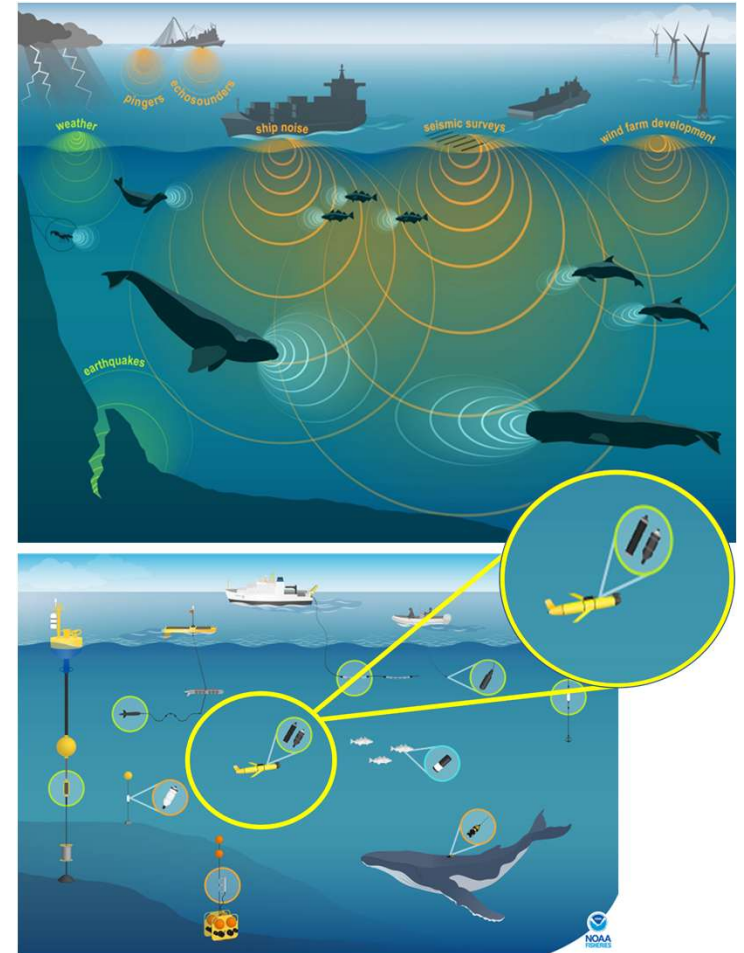
Shannon Rankin

Erin Oleson, Selene Fregosi, Kourtney Burger, Anthony Cossio,
Calla Lloyd-Lim, David K. Mellinger, Erik Norris

NOAA Fisheries UxS-PAM Strategic Initiative

Goal:
PAM-Gliders to augment Traditional Ship Surveys

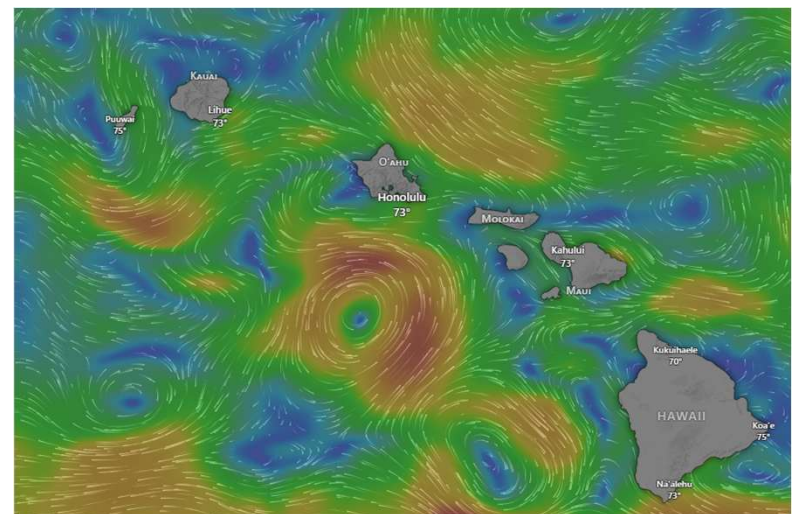
- Build Capacity to sustain glider operations
- Concurrent Glider & Shipboard Surveys
- Develop Spatial Modeling Methods
- In-Water Instrumentation Testing
(Glider Rodeo!)



Objectives

Field testing of PAM-glider platforms to provide data-driven guidance on the capabilities of each system to:

- Work in different operational environments, and
- Address different research questions.



Approach

1



Trimming

Fine-tune adjustments to prepare for rodeo!

2



FAST Mission Mode

Pilot to prioritize trackline progression

3



SLOW Mission Mode

Pilot to minimize noise while reaching end of line

4



INTERMEDIATE Mission Mode

Balance maintaining course w/ reduced self-noise

5



Drift Mode

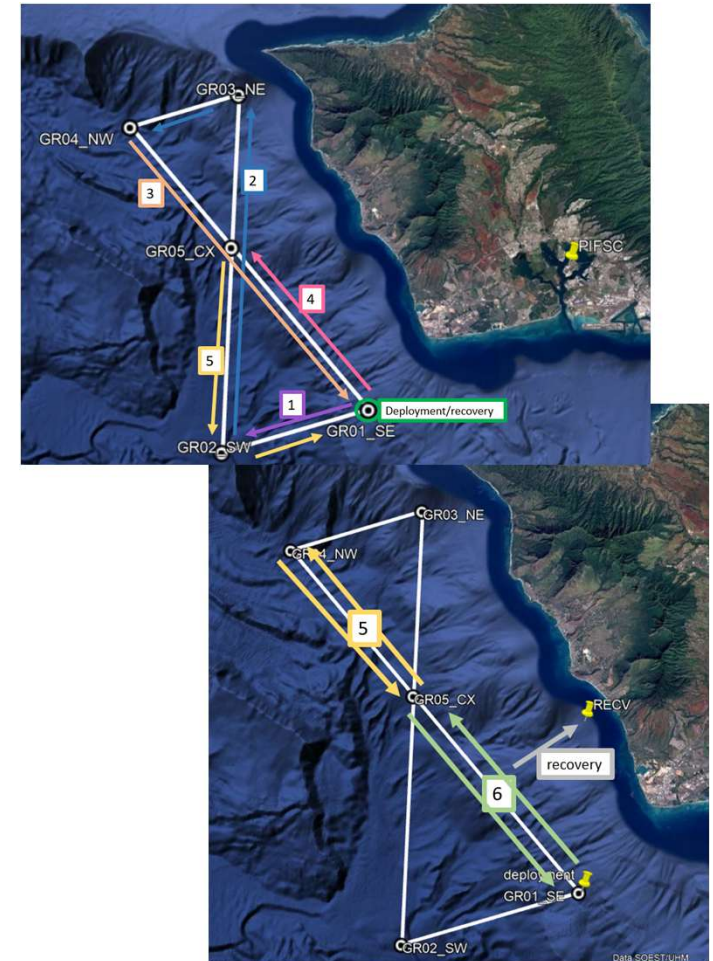
Intermediate mode w/ 20 min drift @ 800m (repeated)

6



Mini Rodeo

Cycle all gliders through Phase 2-5 (12 hrs each)



Assessment Areas

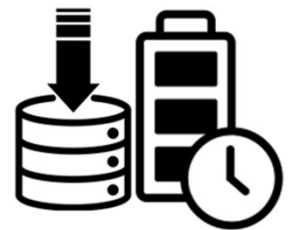


Navigation: Understand navigation strengths & limitations and their impact on endurance, noise, accomplishing mission objectives

Piloting: Understand piloting under different mission modes & their impact on navigation, endurance, noise



Endurance: Assess power and storage endurance under different mission modes



Noise: Assess self-noise and degree to which noise interferes with mission objectives

Rodeo Participants



UW
SeaGlider
WISPR



Hefring
Oceanscout
WISPR



Alseamar
Sea Explorer
Auris



Teledyne-Webb
Slocum
Jasco Ocean Observer (1)
DMON (1)
WISPR (1)



**TELEDYNE
WEBB RESEARCH**
Everywhere you look™

HEFRING
ENGINEERING



JASCO
APPLIED SCIENCES

ALSEAMAR

ALCEN



**Oregon State
University**



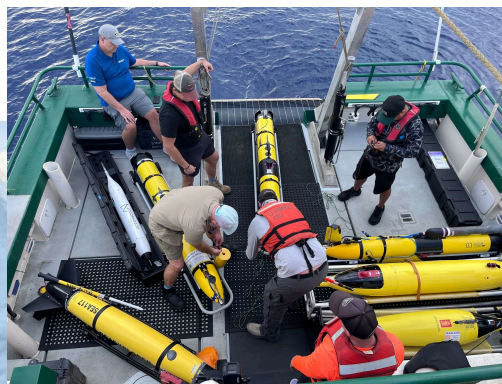
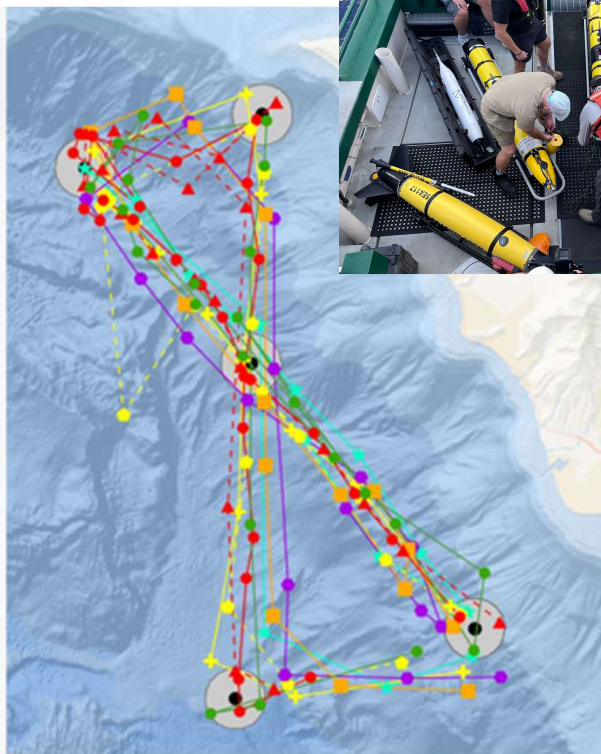
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Fieldwork

Map of Glider Tracks as of 2/10/2026

- Sea Explorer (Alseamar)
- Ocean Scout (Cornell, Belladonna)
- ▲ Ocean Scout (Cornell, Rosie)
- SeaGlider (OSU)
- SeaGlider (PI)
- ✚ Slocum (SW Risso)
- Slocum (SW Stenella)
- ★ Slocum (Teledyne, Jasco)



Plan: 10 Gliders

Deploy: 8 Gliders

Retrieve: 7 Gliders

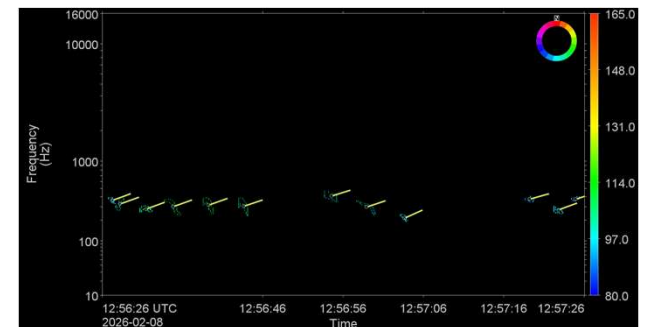
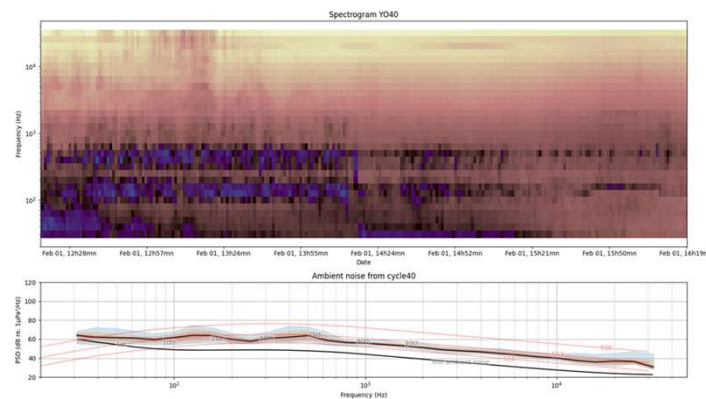
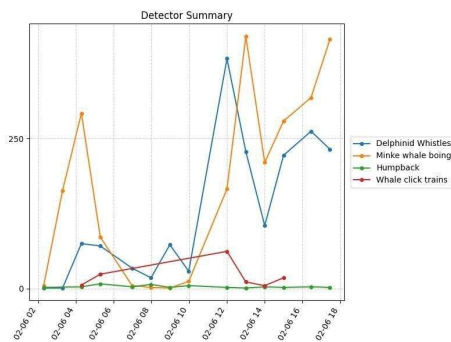
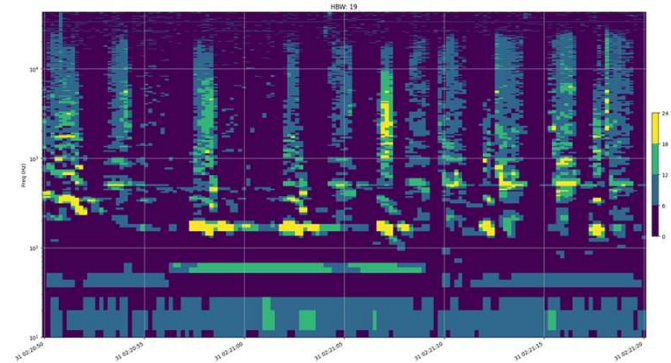
Data: 6.5 Gliders!



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Data!

- 4.5 Tb Audio Data
- 415 km (per glider)



Analysis Questions



How can we pilot gliders to achieve different mission modes?



How is navigation affected by environmental conditions?



How does Mission Mode affect battery endurance/storage capacity?



How does glider noise interfere with study objectives?

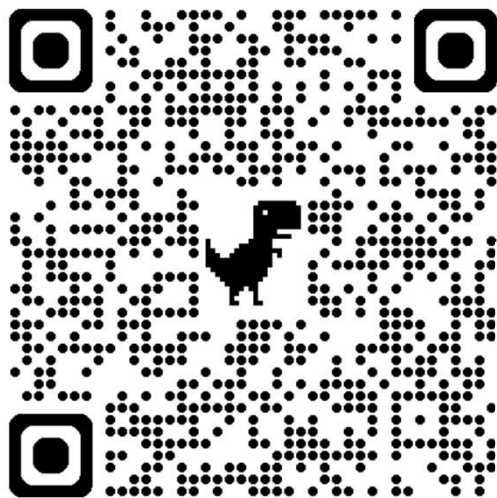


Glider Rodeo Hackweek

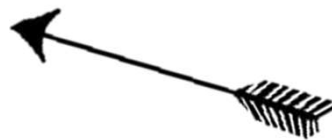
SEPTEMBER 15-18 2026

REMOTE  ONLINE  EVENT !!

NO EXPERIENCE REQUIRED !!



Miss out on the Wet Rodeo?
Join us for the Data Rodeo!



Ideas? Interested? Fill out our survey!



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so many thanks to so many people!

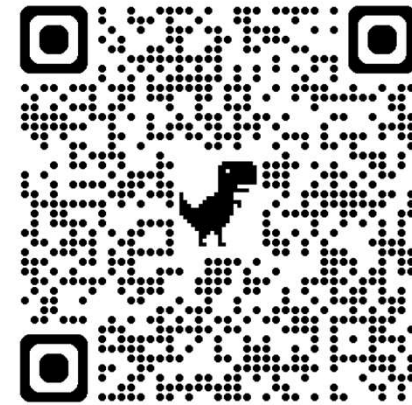


REFOCUS (SWFSC):

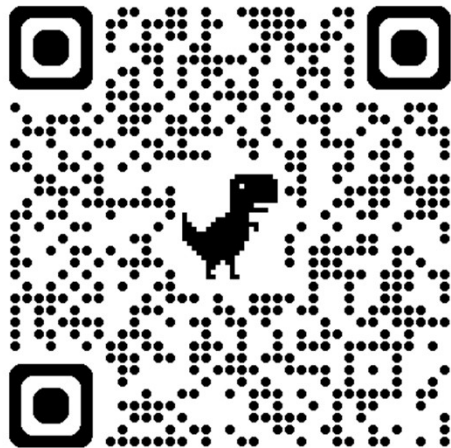
REIMAGINING ECOSYSTEM AND FISHERIES OBSERVATIONS BY COMBINING UxS FLEETS
UxSOC AND REFOCUS PROVIDED SWFSC GLIDERS FOR PAM INTEGRATION



Hackathon Interest Survey



Glider Rodeo Website



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